

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTBS404 - PROBABILITY THEORY AND RANDOM PROCESSES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The addition theorem of probability is used when:
a) Events are mutually exclusive b) Events are independent
c) Events are exhaustive d) None of the above
- 2 If A and B are two independent events, then $P(A \cap B) = ?$
a) $P(A) + P(B)$ b) $P(A) \times P(B)$
c) $P(A) / P(B)$ d) $P(A) - P(B)$
- 3 Which distribution is appropriate for modeling the number of occurrences of an event in a fixed interval of time or space?
a) Binomial distribution b) Poisson distribution
c) Normal distribution d) Exponential distribution
- 4 The mathematical expectation of a discrete random variable X is given by:
a) $\int x f(x) dx$ b) $\sum x P(x)$
c) $\sum x^2 P(x)$ d) $\sum P(x)/x$
- 5 The variance of a random variable is:
a) $E(X)$ b) $E(X^2)$
c) $E[(X - \mu)^2]$ d) $\sqrt{E(X)}$
- 6 The correlation coefficient lies between:
a) 0 and 1 b) -1 and +1
c) $-\infty$ and $+\infty$ d) 0 and ∞
- 7 Spearman's rank correlation is used when:
a) Data is quantitative b) Data is qualitative
c) Data is ordinal d) Data is categorical
- 8 Which of the following best describes regression?
a) Finding the average b) Finding the relationship between variables
c) Testing a hypothesis d) Estimating probability
- 9 If the regression coefficient is 0, then:
a) There is a perfect positive relationship b) There is no linear relationship
c) There is a perfect negative relationship d) The correlation is 1

Q6. Solve Any Two of the following.

- A) What is mean by Testing of hypothesis, level of significance, critical region, Type-I and Type-II errors. 6
- B) A manufacturer claims that the mean breaking strength of safety belts for air 6 passengers produced in his factory is 1275 kg. A sample of 100 belts was tested and the mean breaking strength and Standard Deviation were found to be 1258 and 90 kg respectively. Test the manufacturer's claim at 1% Level of significance.
- C) In two large population there are 30% and 25% respectively of fair haired 6 peoples. Is this difference likely to be hidden in sample of 1200 and 900 respectively from the two populations? Use 5% level of significance.

1714

1714
*** End ***

1714

1714

1714

1714

1714

1714

1714